

RNA-seq recovers a mitochondrial genome that its
own assembly does not annotate: introns, expressed
ORFs and a 2,135-fold gain in *Pleurotus ostreatus*

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Abstract

Mitochondrial genomes are routinely deposited alongside nuclear assemblies and routinely left unannotated, on the reasonable grounds that they are not what the assembly was for. We show what this costs, and that ordinary RNA-seq is sufficient to repair it. The mitochondrial contig of the *Pleurotus ostreatus* BOM_ss5 assembly carries 13 annotated features across 71,949 bp, of which one is protein-coding. Combining tRNAscan-SE, barrnap, ORF prediction under the empirically confirmed genetic code 4 and homology identification, we recover 53 features: 20 protein-coding genes, 25 tRNAs, 2 rRNAs and 6 introns. Read assignment to the mitochondrion rises 2,135-fold, from 343 to 732,325 reads across 16 libraries, and 40 of 47 quantifiable features are detectably expressed. Split-read evidence identifies six spliced introns totalling 4,332 bp, two supported independently by Rfam covariance models for group I introns; the largest is excised at complete efficiency in all 16 libraries. We further show that raw split-read counts are a treacherous measure of splicing: 50 of 56 recurrent junctions splice at under 5% efficiency, and 46 of those lie inside the high-coverage ribosomal RNA region, where they are alignment artefacts rather than introns. Twenty predicted ORFs have no characterised homologue, of which five are expressed in at least half the libraries and cluster at two loci. The mitochondrion is, in this organism, an appreciable and almost entirely undocumented fraction of what RNA-seq actually measures.

Keywords: mitogenome, *Pleurotus ostreatus*, group I intron, genome annotation, mitotranscriptome, RNA-seq

047 **DRAFT**

048

049 All values are generated from the analysis repository and verified by `scripts/32_audit_numbers.py`. Sam-
050 ple provenance and library preparation are described in the companion resource paper. [AUTHORS]
051 markers indicate items only the authors can supply.

051

052 **1 Introduction**

053

054 A fungal genome submission is judged on its nuclear content. The mitochondrial
055 contig is assembled because the reads are there, deposited because completeness is a
056 virtue, and annotated to whatever depth the pipeline happens to reach — which, for
057 organellar sequence, is often barely at all. This is defensible: the assembly was not
058 undertaken to describe the mitochondrion.

059 It is nonetheless consequential for anyone who later maps RNA to that assembly.
060 Fungal mitochondria are transcriptionally active, their genomes are intron-rich [1],
061 and in libraries where ribosomal depletion is incomplete they can account for a sub-
062 stantial share of sequenced material. Reads that map to unannotated mitochondrial
063 sequence are not merely uncounted; they are reported in categories — “no feature”,
064 “ambiguous” — that most workflows summarise and few interrogate.

065 Zaccaron and Bluhm’s analysis of the *Bipolaris sorokiniana* mitogenome [1]
066 describes what a properly annotated fungal mitochondrion contains: protein-coding
067 genes interrupted by numerous group I and group II introns, a full tRNA complement,
068 and dozens of free-standing ORFs of unknown function. We asked how much of that
069 is recoverable for *Pleurotus ostreatus* from RNA-seq alone, using libraries generated
070 for an entirely different purpose.

071

072 **2 Results**

073

074 **2.1 The deposited mitogenome is almost unannotated**

075

076 The mitochondrial contig CM148777.1 of the BOM_ss5 assembly is 71,949 bp with
077 26.6% GC. Its GenBank annotation carries **13 gene features: 12 tRNAs and**
078 **one protein-coding gene** (Table 1). A typical fungal mitogenome encodes roughly
079 fourteen proteins, two ribosomal RNAs and twenty-five tRNAs, so essentially the
080 entire protein-coding and ribosomal complement is missing.

081

082 **2.2 A corrected annotation from four independent lines of** 083 **evidence**

084

085 We first established the genetic code empirically rather than assuming it. EMBOSS
086 `getorf` yields 51 ORFs of at least 100 codons under translation table 4 against 34
087 under table 1, confirming that TGA encodes tryptophan — the mould mitochondrial
088 code.

089 Combining tRNAscan-SE (25 tRNAs, corroborated by aragorn at 22), barnnap for
090 ribosomal RNA, and ORF prediction with DIAMOND identification against Swiss-
091 Prot gives **53 features: 20 protein-coding, 25 tRNA, 2 rRNA and 6 introns**
092 (Fig. 1, Table 1). Identified products include cytochrome b, cytochrome c oxidase

subunits 2 and 3, NADH dehydrogenase subunits 2, 4 and 5, ATP synthase subunit a, and the small ribosomal subunit protein uS3m. Two products are intron-encoded endonucleases of the aI5 and aI8 families, which is itself evidence of intron content: these enzymes are encoded *within* group I introns.

Every feature in the released GFF carries the evidence that produced it, so a reader who distrusts any one line can discount it without re-deriving the rest.

2.3 Read assignment improves 2,135-fold

Across all 16 libraries, 21,180,726 alignments fall on the mitochondrion. The stock annotation assigns **343** of them. The corrected annotation assigns **732,325**, a 2,135-fold increase, and **40 of 47 quantifiable features are detectably expressed** (Fig. 3a).

The corrected figure is still only 3.5% of mitochondrial reads, and the reason is instructive: barrnap identifies the conserved cores of the ribosomal RNA genes, a few hundred base pairs each, whereas the transcribed ribosomal regions span several kilobases. Annotation completeness is not binary, and the residue here is ribosomal rather than protein-coding.

2.4 Six introns, confirmed by splicing

Split-read junctions were pooled across all 16 libraries and filtered to those seen in at least three libraries with at least ten supporting reads, leaving 56 recurrent junctions of 459 raw.

Six of these splice at greater than 50% efficiency and we regard them as genuine introns (Table 2, Fig. 2). They total 4,332 bp, 6.0% of the mitogenome. The largest, at 6,796–7,232, is supported by 4,782 reads, is present in **all 16 libraries**, and is excised at **complete efficiency in every one**. Two of the six coincide with Rfam covariance-model predictions for group I introns, one at $E = 2.7 \times 10^{-15}$.

Covariance-model search independently identifies nine group I intron candidates on the mitogenome and no significant group II candidate. This matters for interpretation: group II introns encode reverse transcriptases ancestral to telomerase and to non-LTR retrotransposon RTs, whereas group I introns encode homing endonucleases and carry no such connection. The intron-encoded endonucleases recovered above are consistent with an exclusively group I complement, and no inference about retroelement or telomerase biology is available from these data.

2.5 Most recurrent junctions are artefacts, and they are not randomly placed

The remaining 50 junctions splice at under 5% efficiency: for each, the overwhelming majority of reads crossing the donor site do so without a gap. **Forty-six of those 50 lie inside the mitochondrial ribosomal RNA block** (Fig. 2b), where coverage reaches 10^6 and a small per-read misalignment rate produces a large absolute number of spurious gapped alignments.

This has a practical consequence we did not anticipate. Raw split-read counts on the mitochondrion vary 20–34-fold *between replicates of the same tissue*, which invites

139 the interpretation that splicing is condition-dependent. It is not: that variance is
140 carried almost entirely by the artefactual junctions, while the genuine introns splice to
141 completion uniformly. **Splicing efficiency, not split-read count, is the quantity**
142 **that behaves.** We accordingly make no claim about differential splicing between
143 tissues, and would caution against such claims from comparable data without an
144 efficiency denominator.

145

146 2.6 Expressed ORFs without characterised homologues

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148 Twenty of the 51 predicted ORFs have no DIAMOND hit to Swiss-Prot and do not
149 overlap an annotated feature. Six are detected at five or more reads in at least half
150 the libraries; one lies inside a ribosomal RNA block and is discarded because its count
151 reflects ribosomal rather than ORF transcription, leaving **five credible expressed**
152 **candidates** (Table 3, Fig. 3b). They fall at two loci, near 32.8kb and 65.2kb, with
153 overlapping predictions at each, so the number of distinct transcribed units is probably
154 two rather than five.

155 Absence of a Swiss-Prot hit is weak evidence of novelty — Swiss-Prot covers 41%
156 of this organism’s nuclear proteome — so these are candidates for follow-up, not
157 discoveries.

158

159 3 Discussion

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161 Three points generalise.

162 **Organellar annotation is a systematic blind spot with a measurable cost.**

163 The gap here is not subtle: one annotated protein-coding gene where there are twenty,
164 and a 2,135-fold difference in what can be counted. Any study mapping RNA to a
165 fungal assembly inherits this, and inherits it silently, because unannotated sequence
166 produces no error.

167 **RNA-seq is sufficient to repair it.** Nothing here required mitochondrial enrich-
168 ment, long reads, or a dedicated experiment. The libraries were prepared for a tissue
169 comparison and the mitochondrial signal was, from that perspective, contamination.
170 Where poly(A) selection is incomplete — common enough — the resulting
171 non-polyadenylated material is exactly what makes organellar transcript annotation
172 possible.

173 **Split-read counts are not splicing measurements.** The distinction between
174 a junction and an efficiently spliced junction is the difference between 56 apparent
175 introns and six real ones, and between an apparent 20–34-fold biological effect and
176 an artefact of coverage. We would not have separated them without computing the
177 unspliced denominator.

178

179 3.1 Limitations

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181 The annotation is computational throughout and no gene model has been verified
182 by RACE, proteomics or targeted sequencing. Protein identifications rest on homol-
183 ogy, and the ORF predictions inherit whatever errors are present in the underlying
184 assembly, which is contig-level. Coverage is uneven — median depth 2 outside the

ribosomal regions — so absence of expression is weak evidence for any individual feature. The intron catalogue is limited to introns that are spliced in these libraries; a constitutively retained intron is invisible to this approach by construction. Rfam group I models are sensitive but not specific at the thresholds used, and seven of the nine covariance-model hits lack independent split-read support.

4 Methods

Culture, RNA extraction, sequencing and read processing are described in the companion resource paper. In brief, 16 libraries were sequenced as poly(A)-selected 3'-tag libraries and aligned with HISAT2 [2] to *P. ostreatus* BOM_ss5 (GCA_056149245.1); alignments were reduced to primary records before analysis.

Annotation. Genetic code determined by comparing EMBOSS `getorf` yield under tables 1 and 4. tRNAs from tRNAscan-SE [3] in organellar mode, cross-checked with aragorn [4]; rRNAs from barrnap [5]; protein-coding genes from `getorf` ORFs identified by DIAMOND [6] blastp against Swiss-Prot [7], retaining the longest ORF where predictions nested. Reconciliation and GFF output by `scripts/40_mito_annotate.py`.

Introns. Junctions extracted from CIGAR N operations, pooled across libraries and filtered to ≥ 3 libraries and ≥ 10 reads. Splicing efficiency computed per library as spliced reads over spliced plus donor-spanning unspliced reads, requiring at least five informative reads. Group I/II predictions from Infernal [8] `cmsearch` against Rfam [9] models RF00028, RF00029, RF01998, RF02001 and RF02003.

Quantification. `featureCounts` [10] against the corrected annotation and, for comparison, against the stock GenBank features.

Figures

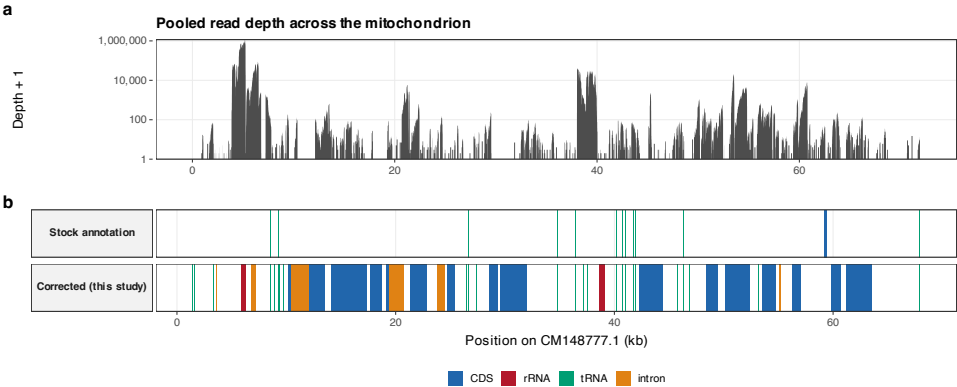


Fig. 1 The deposited mitogenome is almost unannotated. **a**, Pooled read depth across CM148777.1. **b**, Annotated features: the stock GenBank annotation above, the corrected annotation below.

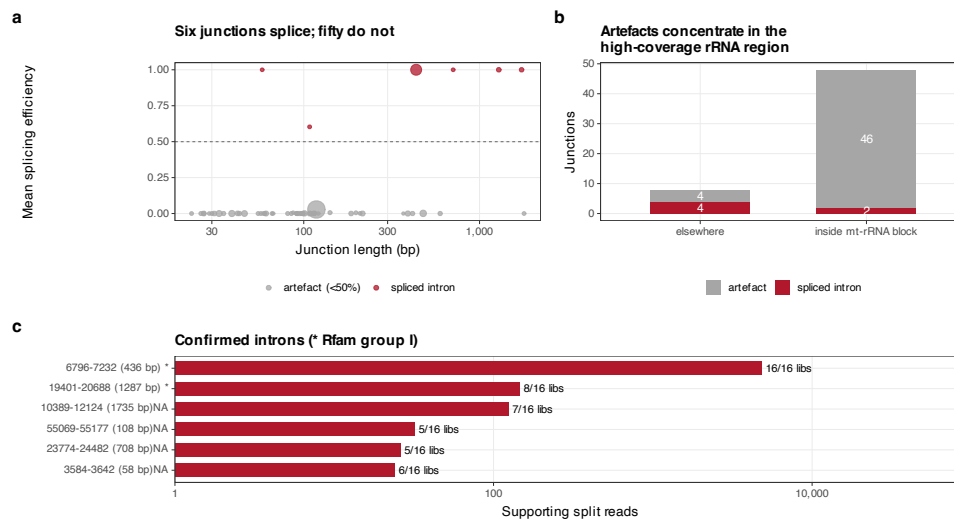


Fig. 2 Six introns among fifty-six recurrent junctions. **a**, Splicing efficiency against junction length; point size is supporting reads. **b**, Artefactual junctions concentrate inside the ribosomal RNA block. **c**, The six confirmed introns; asterisks mark Rfam group I support.

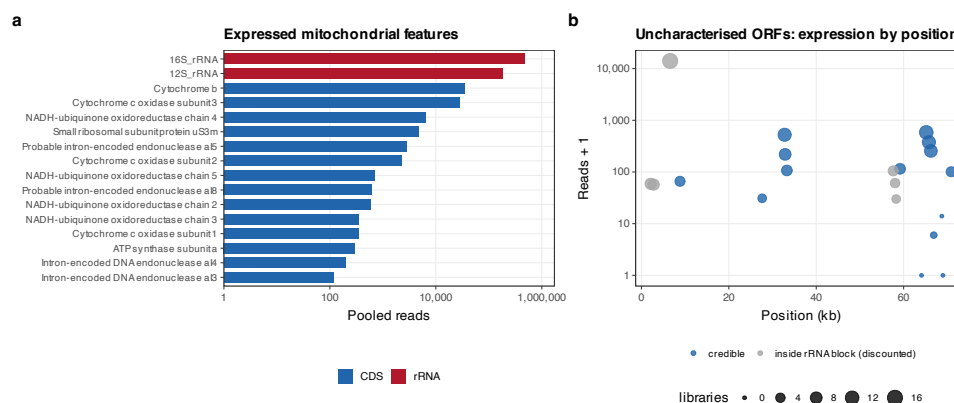


Fig. 3 Expressed mitochondrial features and candidate ORFs. **a**, Pooled reads for the most highly expressed annotated features. **b**, Uncharacterised ORFs by position; those inside ribosomal RNA blocks are discounted.

Tables

Table 1 Annotation of the *P. ostreatus* BOM_ss5 mitochondrion (CM148777.1, 71,949 bp).

Feature class	Stock GenBank	Corrected (this study)
Protein-coding (CDS)	1	20
Ribosomal RNA	0	2
Transfer RNA	12	25
Introns	0	6
Total	13	53

Corrected features derive from tRNAscan-SE, barnap, EMBOSS `getorf` under genetic code 4 with DIAMOND identification, and split-read intron evidence.

Table 2 Mitochondrial introns confirmed by split-read evidence (splicing efficiency >0.5).

Start	End	Length	Reads	Libraries	Efficiency	Rfam model
6796	7232	436	4,782	16/16	1.0	group I (E=2.7e-15)
19401	20688	1287	146	8/16	1.0	group I (E=1.2e-13)
10389	12124	1735	124	7/16	1.0	–
55069	55177	108	32	5/16	0.603	–
23774	24482	708	26	5/16	1.0	–
3584	3642	58	24	6/16	1.0	–

Efficiency is spliced reads divided by spliced plus donor-spanning unspliced reads, averaged over libraries with at least five informative reads.

Table 3 Expressed mitochondrial ORFs with no Swiss-Prot homologue, outside the rRNA blocks.

ORF	Start	End	Length (nt)	Strand	Reads	Libraries
CM148777.1_27	65174	66319	1146	+	584	12/16
CM148777.1_14	32764	33906	1143	+	524	11/16
CM148777.1_33	65782	66177	396	-	376	11/16
CM148777.1_32	66224	66730	507	-	254	10/16
CM148777.1_44	32880	33218	339	-	218	8/16

One further ORF passed the expression filter but lies inside an mt-rRNA block and is excluded: its read count reflects ribosomal RNA rather than ORF transcription.

Data availability. https://github.com/dr-richard-barker/Myco_tissue_RNAseq. The corrected annotation is provided as `refs/BOM_ss5/mitogenome.gff`, with the intron catalogue and ORF tables alongside.

Author contributions. [AUTHORS: CRediT statement for all eight authors.]

Funding.	[AUTHORS.]	369
Competing interests.	[AUTHORS: declaration.]	370
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		372
References		373
		374
[1]	Zaccaron, A. Z. & Bluhm, B. H. Erosion of genome sequences and emergence of novel genes in the mitochondrial genome of the fungal pathogen <i>bipolaris sorokiniana</i> . <i>Frontiers in Microbiology</i> 11 , 863 (2020).	375
		376
		377
[2]	Kim, D., Paggi, J. M., Park, C., Bennett, C. & Salzberg, S. L. Graph-based genome alignment and genotyping with hisat2 and hisat-genotype. <i>Nature Biotechnology</i> 37 , 907–915 (2019).	378
		379
		380
		381
[3]	Chan, P. P., Lin, B. Y., Mak, A. J. & Lowe, T. M. trnascan-se 2.0: improved detection and functional classification of transfer rna genes. <i>Nucleic Acids Research</i> 49 , 9077–9096 (2021).	382
		383
		384
		385
[4]	Laslett, D. & Canback, B. Aragorn, a program to detect trna genes and tmrna genes in nucleotide sequences. <i>Nucleic Acids Research</i> 32 , 11–16 (2004).	386
		387
		388
[5]	Seemann, T. barrnap: Basic rapid ribosomal rna predictor. https://github.com/tseemann/barrnap (2013).	389
		390
		391
[6]	Buchfink, B., Xie, C. & Huson, D. H. Fast and sensitive protein alignment using diamond. <i>Nature Methods</i> 12 , 59–60 (2015).	392
		393
		394
[7]	The UniProt Consortium. Uniprot: the universal protein knowledgebase in 2023. <i>Nucleic Acids Research</i> 51 , D523–D531 (2023).	395
		396
		397
[8]	Nawrocki, E. P. & Eddy, S. R. Infernal 1.1: 100-fold faster rna homology searches. <i>Bioinformatics</i> 29 , 2933–2935 (2013).	398
		399
		400
[9]	Kalvari, I., Nawrocki, E. P., Ontiveros-Palacios, N. <i>et al.</i> Rfam 14: expanded coverage of metagenomic, viral and microrna families. <i>Nucleic Acids Research</i> 49 , D192–D200 (2021).	401
		402
		403
[10]	Liao, Y., Smyth, G. K. & Shi, W. featurecounts: an efficient general purpose program for assigning sequence reads to genomic features. <i>Bioinformatics</i> 30 , 923–930 (2014).	404
		405
		406
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		409
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		411
		412
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		414